← Back

Knowledge Check: Assessment and Applications
Practice Assignment • 15 min

Due Oct 19, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Which **measures** evaluate a partition by first computing a value **per community**, then averaging over **all** communities? (Select **all** correct answers.) 1 / 1 point

- [x] Density

  ✓ Nice work

  Correct. Density is computed per community (fraction of possible edges used), then averaged.

- [ ] Performance
- [ ] Coverage
- [x] Separability

  ✓ Nice work

  Correct. Separability calculates a ratio for each community, then averages.

2. **Performance** rewards partitions with **many** intra-community edges and: 1 / 1 point

- (•) Many **inter-community non-edges**, i.e., node pairs in different communities that are not connected
- ( ) Randomly assigned nodes to ensure coverage is high
- ( ) A high density in every single community
- ( ) A high total edge count across the entire network

  ✓ Nice work

  Correct! Performance counts intra-community edges + inter-community non-edges in its numerator.

3. **Separability** measures a community's ratio of: 1 / 1 point

- ( ) The community's total nodes to the entire network's total nodes
- (•) **Intra-community edges** to **inter-community edges** for each community, then averages across all communities
- ( ) Edges to non-edges within the same group
- ( ) Node degrees to total edges in the network

  ✓ Nice work

  Correct! Separability = (within-community edges) / (edges to outside) for each community, then averaged.